

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Cleaning Challenge

Course name :Query Processing for Data Science

Course Code: Dsa0503

COURSE OUTCOME COVERED

CO3: Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex.

Assessment Tool Weightage: 35%

Total Marks: 20

Code of Conduct

I_V.Tejaswini(192424360) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Criteria	Anomaly Detection (4 Marks)	Cleaning Methods (4 Marks)	Data Transformation (4 Marks)	Analytical Comparison (4 Marks)	Governance Discussion (4 Marks)
Marks					

Total Marks Awarded Questions:

Question 1: E-Commerce Customer Transactions Dataset

An online retail company has collected customer transaction data from multiple sales channels. The management wants to identify the top-performing customer segments and products. However, the dataset contains several quality issues.

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C101	Raj Kumar	Male	28	Electronics	2	15000 INR	10-01-2025	Chennai
C102	Priya S	Female	35	Fashion	3	1200 INR	11-01-2025	chennai
C103	Arun V	Male	150	Electronics	1	300 USD	12-01-2025	CHENNAI
C104	Kavya R	Female	0	Grocery	-2	800 INR	15-01-2025	Coimbatore
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
NULL	Sneha M	Female	30	Electronics	2	450 USD	05-02-2025	Trichy
C107	Deepak N	Male		Grocery	3	600 INR	07-02-2025	Tiruchirappalli
C108	Meena K	Female	27	Fashion	1	1800 INR	2025-13-01	Salem
C109	John D	Male	31	Electronics	4	250 EUR	15-02-2025	Chennai

Data Issues

- Missing Customer IDs
- Duplicate transactions
- Negative quantities
- Unit prices recorded in different currencies
- Age values such as 0, 150, and blanks

- Inconsistent city names (e.g., Chennai, chennai, CHENNAI)
- Incorrect purchase dates

Tasks

1. Identify and classify all data quality issues. (4 Marks)
2. Design a cleaning strategy for each issue. (4 Marks)
3. Perform data standardization and transformation. (4 Marks)
4. Analyze how cleaning affects total revenue calculations. (4 Marks)
5. Generate a report discussing business risks if the data were not cleaned (4 Marks)

code:

```
import pandas as pd

df = pd.DataFrame({
    "ID": ["C101", "C102", "C103", "C104", "C105", "C105", None, "C107", "C108", "C109"],
    "Age": [28, 35, 150, 0, 42, 42, 30, None, 27, 31],
    "Qty": [2, 3, 1, -2, 5, 5, 2, 3, 1, 4],
    "Price": ["15000 INR", "1200 INR", "300 USD", "800 INR", "2500 INR",
              "2500 INR", "450 USD", "600 INR", "1800 INR", "250 EUR"],
    "City": ["Chennai", "chennai", "CHENNAI", "Coimbatore", "Madurai", "Madurai",
             "Trichy", "Tiruchirappalli", "Salem", "Chennai"]
})

df = df.drop_duplicates()
df["ID"] = df["ID"].fillna("Unknown")
df.loc[(df.Age <= 0) | (df.Age > 120), "Age"] = df.Age.median()
df.loc[df.Qty < 0, "Qty"] = 0

df["City"] = df.City.str.title().replace({"Tiruchirappalli": "Trichy"})

rate = {"INR": 1, "USD": 83, "EUR": 90}
df["Price"] = df.Price.apply(lambda x: float(x.split()[0]) * rate[x.split()[1]])
df["Revenue"] = df.Qty * df.Price

print(df)
print("Total Revenue:", df.Revenue.sum())
print("\nRevenue by Category/City:")
print(df.groupby("City").Revenue.sum())
```

output:

```
...      ID  Age  Qty   Price      City  Revenue
0      C101 28.0   2  15000.0   Chennai  30000.0
1      C102 35.0   3   1200.0   Chennai   3600.0
2      C103 30.5   1  24900.0   Chennai  24900.0
3      C104 30.5   0    800.0  Coimbatore    0.0
4      C105 42.0   5   2500.0   Madurai  12500.0
6   Unknown 30.0   2  37350.0   Trichy   74700.0
7      C107  NaN   3    600.0   Trichy   1800.0
8      C108 27.0   1   1800.0   Salem   1800.0
9      C109 31.0   4  22500.0   Chennai  90000.0
Total Revenue: 239300.0

Revenue by Category/City:
City
Chennai      148500.0
Coimbatore      0.0
Madurai      12500.0
Salem          1800.0
Trichy        76500.0
Name: Revenue, dtype: float64
```

Question 2: Hospital Patient Records Dataset

A hospital wants to analyze patient admissions and treatment effectiveness. Data were entered manually by different departments, leading to inconsistencies.

PatientID	Name	Age	Gender	AdmissionDate	DischargeDate	Diagnosis	TreatmentCost	DoctorID
P1001	Ravi Kumar	45	M	10-01-2025	15-01-2025	Diabetes	25000	D101
P1002	Priya S	38	F	12-01-2025	11-01-2025	Hypertension	18000	D102
P1003	Arun V	125	M	13-01-2025	20-01-2025	diabetes	22000	D101

NULL	Kavya R	29	F	15-01-2025	17-01-2025	Diabetes	15000	D103
P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1006	Sneha M	34	F	15-01-2025	18-01-2025	Hypertension	12000	D102
P1007	Deepak N		M	01-02-2025	05-02-2025	Asthma	10000	D104
P1008	Meena K	26	F	10-02-2025	15-02-2025	HYPERTENSION	13500	D102
P1009	John D	40	M	12-02-2025	18-02-2025	Diabetes	20000	D101

Data Issues

- Missing patient IDs
- Duplicate patient records
- Discharge dates before admission dates
- Null treatment costs
- Inconsistent diagnosis names
- Multiple formats of dates
- Unrealistic patient ages (>120)

Tasks

1. Detect all inconsistencies and anomalies. (4 Marks)
2. Develop rules for handling missing and invalid values. (4 Marks)
3. Create a process to standardize diagnosis names. (4 Marks)

4. Evaluate how cleaning changes average treatment cost and hospital stay duration. (4 Marks)
5. Recommend data governance measures to prevent future errors. (4 Marks)

Code:

```
import pandas as pd

df=pd.DataFrame({
    "ID":["P1001","P1002","P1003",None,"P1005","P1006","P1007","P1008","P1009"],
    "Age":[45,38,125,29,51,51,34,None,26,40],
    "Adm":["10-01-2025","12-01-2025","13-01-2025","15-01-2025","18-01-2025",
    "18-01-2025","15-01-2025","01-02-2025","10-02-2025","12-02-2025"],
    "Dis":["15-01-2025","11-01-2025","20-01-2025","17-01-2025","25-01-2025",
    "25-01-2025","18-01-2025","05-02-2025","15-02-2025","18-02-2025"],
    "Diagnosis":["Diabetes","Hypertension","diabetes","Diabtes","Asthma","Asthma",
    "Hypertension","Asthma","HYPERTENSION","Diabetes"],
    "Cost":[25000,18000,22000,15000,None,None,12000,10000,13500,20000]
})

df=df.drop_duplicates()
df.ID=df.ID.fillna("Unknown")
df.loc[df.Age>120,"Age"]=df.Age.median()

df.Diagnosis=df.Diagnosis.str.lower().replace({"diabtes":"diabetes"})
df.Diagnosis=df.Diagnosis.str.title()

df.Adm=pd.to_datetime(df.Adm,dayfirst=True)
df.Dis=pd.to_datetime(df.Dis,dayfirst=True)
df.loc[df.Dis<df.Adm,"Dis"]=None

df.Cost=df.Cost.fillna(df.Cost.median())
df["Stay"]=(df.Dis-df.Adm).dt.days

print(df)
print("Average Cost:",df.Cost.mean())
print("Average Stay:",df.Stay.mean())
```

Output:

...	ID	Age	Adm	Dis	Diagnosis	Cost	Stay
0	P1001	45.0	2025-01-10	2025-01-15	Diabetes	25000.0	5.0
1	P1002	38.0	2025-01-12	NaT	Hypertension	18000.0	NaN
2	P1003	39.0	2025-01-13	2025-01-20	Diabetes	22000.0	7.0
3	Unknown	29.0	2025-01-15	2025-01-17	Diabetes	15000.0	2.0
4	P1005	51.0	2025-01-18	2025-01-25	Asthma	16500.0	7.0
6	P1006	34.0	2025-01-15	2025-01-18	Hypertension	12000.0	3.0
7	P1007	NaN	2025-02-01	2025-02-05	Asthma	10000.0	4.0
8	P1008	26.0	2025-02-10	2025-02-15	Hypertension	13500.0	5.0
9	P1009	40.0	2025-02-12	2025-02-18	Diabetes	20000.0	6.0
Average Cost: 16888.88888888889							
Average Stay: 4.875							

Question 3: Banking Loan Approval Dataset

A bank uses customer data to determine loan eligibility. The dataset contains errors that may affect risk assessment and approval decisions.

Custom erID	AnnualIn come	CreditS core	LoanAm ount	Employment Status	A ge	Existing Debt	LoanSt atus
L001	750000	720	300000	Full Time	35	50000	Approv ed
L002	900K	680	400000	FT	42	100000	Approv ed
L003	1200000	NULL	600000	fulltime	28	200000	Approv ed
L004	150000	450	-200000	Part Time	30	50000	Rejecte d
L005	5000000	790	1000000	Self Employed	16	10000	Approv ed
L006	450000	350	250000	Unemployed	40	300000	Rejecte d
L007	75000000	800	500000	Full Time	45	100000	Approv ed
L008	650000	690	350000	FT	33	70000	Approv ed
L008	650000	690	350000	FT	33	70000	Approv ed
L010	NULL	710	300000	Full Time	37	80000	Approv ed

Data Issues

- Missing credit scores
- Income values recorded in different units

- Negative loan amounts
- Duplicate customer IDs
- Age values below 18
- Employment status entered differently (FT, Full Time, fulltime)
- Extreme outliers in income

Tasks

1. Identify critical issues that could impact loan decisions. (4 Marks)
2. Propose cleaning techniques for missing and inconsistent data. (4 Marks)
3. Detect and justify treatment of outliers. (4 Marks)
4. Analyze the effect of cleaning on loan approval rates. (4 Marks)
5. Discuss ethical implications of incorrect data cleaning decisions. (4 Marks)

Code

```
import pandas as pd

df=pd.DataFrame({
    "ID":["L001","L002","L003","L004","L005","L006","L007","L008","L008","L010"],
    "Income":["750000","900K","1200000","150000","5000000","450000","75000000","650000","650000",None],
    "Credit":[720,680,None,450,790,350,800,690,690,710],
    "Loan":[300000,400000,600000,-200000,1000000,250000,500000,350000,350000,300000],
    "Job":["Full Time","FT","fulltime","Part Time","Self Employed","Unemployed",
    "Full Time","FT","FT","Full Time"],
    "Age":[35,42,28,30,16,40,45,33,33,37],
    "Status":["Approved","Approved","Approved","Rejected","Approved","Rejected",
    "Approved","Approved","Approved","Approved"]
})

df=df.drop_duplicates()

df.Income=df.Income.apply(
lambda x: float(str(x).replace("K","000")) if pd.notna(x) else None)
df.Income=df.Income.fillna(df.Income.median())

df.Credit=df.Credit.fillna(df.Credit.median())

df.loc[df.Loan<0,"Loan"]=df.Loan[df.Loan>0].median()
df.loc[df.Age<18,"Age"]=df.Age.median()

df.Job=df.Job.replace({"FT":"Full Time","fulltime":"Full Time"})

print(df)
print("\nLoan Approval Rate:",
      round((df.Status=="Approved").mean()*100,2),"%")
```


Output:

```
print(mean_approval_rate)
round((df.Status=="Approved").mean()*100,2),"%")

...
ID      Income  Credit   Loan      Job   Age   Status
0  L001    750000.0    720.0  300000    Full Time  35  Approved
1  L002    900000.0    680.0  400000    Full Time  42  Approved
2  L003   1200000.0    700.0  600000    Full Time  28  Approved
3  L004    150000.0    450.0  375000    Part Time  30  Rejected
4  L005   5000000.0    790.0 1000000    Self Employed  35  Approved
5  L006    450000.0    350.0  250000    Unemployed  40  Rejected
6  L007   7500000.0    800.0  500000    Full Time  45  Approved
7  L008    650000.0    690.0  350000    Full Time  33  Approved
9  L010    825000.0    710.0  300000    Full Time  37  Approved

Loan Approval Rate: 77.78 %
```

Question 4: Smart City Traffic Monitoring Dataset

A smart city project collects traffic sensor data to optimize traffic signals. Sensor malfunctions and transmission errors have affected data quality.

SensorID	Timestamp	VehicleCount	AverageSpeed	TrafficSignalStatus	RoadName	WeatherCondition
S101	01-01-2025 08:00	120	45	Green	Anna Salai	Sunny
S102	01-01-2025 08:05	135	52	Red	anna salai	Sunny
S103	01-01-2025 08:10	-50	48	Green	ANNA SALAI	Cloudy
S104	01-01-2025 08:15	140	350	Yellow	GST Road	Rainy
S105	NULL	110	42	Green	GST ROAD	Sunny
S106	01-01-2025 08:25	130	46	Red	OMR	NULL

S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S107	01-01-2025 08:30	150	55	Green	Old Mahabalipuram Road	Sunny
S10A	01-01-2025 08:35	145	53	Green	OMR	Cloudy
S109	01-01-2025 08:40	138	50	Green	GST Road	Rainy

Data Issues

- Missing timestamps
- Duplicate sensor readings
- Vehicle counts recorded as negative numbers
- Average speed exceeding 300 km/h
- Missing weather information
- Inconsistent road naming conventions
- Sensor IDs with formatting errors

Tasks

1. Assess the quality of the traffic dataset. (4 Marks)
2. Recommend cleaning techniques for time-series data. (4 Marks)
3. Design rules to identify faulty sensor readings. (4 Marks)
4. Evaluate how cleaning affects congestion analysis. (4 Marks)
5. Suggest automated monitoring techniques for future data collection. (4 Marks)

Code:

```
import pandas as pd

df=pd.DataFrame({
    "ID":["S101","S102","S103","S104","S105","S106","S106","S107","S10A","S109"],
    "Time":["01-01-2025 08:00","01-01-2025 08:05","01-01-2025 08:10",
    "01-01-2025 08:15",None,"01-01-2025 08:25","01-01-2025 08:25",
    "01-01-2025 08:30","01-01-2025 08:35","01-01-2025 08:40"],
    "Vehicles":[120,135,-50,140,110,130,130,150,145,138],
    "Speed":[45,52,48,350,42,46,46,55,53,50],
    "Road":["Anna Salai","anna salai","ANNA SALAI","GST Road","GST ROAD",
    "OMR","OMR","Old Mahabalipuram Road","OMR","GST Road"]
})

df=df.drop_duplicates()
df.Time=pd.to_datetime(df.Time,dayfirst=True).ffill()

df.loc[df.Vehicles<0,"Vehicles"]=None
df.loc[df.Speed>300,"Speed"]=None

df.Vehicles=df.Vehicles.interpolate()
df.Speed=df.Speed.interpolate()

df.Road=df.Road.str.upper().replace({
    "OLD MAHABALIPURAM ROAD":"OMR"})

df["Congestion"]=pd.cut(
    df.Speed,[0,40,50,100],
    labels=["High","Medium","Low"])

print(df)
print("\nCongestion:")
print(df.Congestion.value_counts())
```

...	ID	Time	Vehicles	Speed	Road	Congestion
0	S101	2025-01-01 08:00:00	120.0	45.0	ANNA SALAI	Medium
1	S102	2025-01-01 08:05:00	135.0	52.0	ANNA SALAI	Low
2	S103	2025-01-01 08:10:00	137.5	48.0	ANNA SALAI	Medium
3	S104	2025-01-01 08:15:00	140.0	45.0	GST ROAD	Medium
4	S105	2025-01-01 08:15:00	110.0	42.0	GST ROAD	Medium
5	S106	2025-01-01 08:25:00	130.0	46.0	OMR	Medium
7	S107	2025-01-01 08:30:00	150.0	55.0	OMR	Low
8	S10A	2025-01-01 08:35:00	145.0	53.0	OMR	Low
9	S109	2025-01-01 08:40:00	138.0	50.0	GST ROAD	Medium

```
Congestion:
Congestion
Medium    6
Low       3
High      0
Name: count, dtype: int64
```

Question 5: Global Sales Performance Dataset

A multinational company has merged sales data from five countries. Management wants to evaluate regional performance and forecast future sales. The merged dataset contains severe data quality problems.

Region	Country	SalesPerson	Product	SalesAmount	Currency	SalesDate	CustomerRating
Asia	India	Raj Kumar	Laptop	50000	INR	10-01-2025	4
Asia	India	Rajkumar	Laptop	55000	INR	10-01-2025	5
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
North America	USA	John D	Laptop	1500	USD	01-15-2025	6
Asia	India	Priya S	Mobile	NULL	INR	18-01-2025	3
Europe	France	Pierre L	Ordinateur Portable	1300	EUR	20-01-2025	4
Asia	India	Rahul P	Mobile	9999999	INR	25-01-2025	5
UK	UK	David T	Laptop	1100	GBP	28-01-2025	2
Asia	India	Priya S	Smartphone	35000	INR	30-01-2025	0

Data Issues

- Multiple currencies (USD, EUR, INR, GBP)
- Missing sales amounts

- Duplicate regional entries
- Customer ratings outside the scale (1–5)
- Different date formats across countries
- Product names in multiple languages
- Salespersons recorded with spelling variations
- Extreme sales outliers due to data-entry errors

Tasks

1. Develop a comprehensive data-cleaning framework for multinational datasets. (4 Marks)
2. Design methods to standardize currencies, products, and dates. (4 Marks)
3. Handle missing values and outliers using appropriate analytical techniques. (4 Marks)
4. Compare regional sales rankings before and after cleaning. (4 Marks)
5. Prepare an executive report explaining how data quality influences strategic business decisions. (4 Marks)

Code:

```
import pandas as pd

df=pd.DataFrame({
    "Region":["Asia","Asia","Europe","Europe","North America",
    "Asia","Europe","Asia","UK","Asia"],
    "SalesPerson":["Raj Kumar","Rajkumar","Anna M","Anna M","John D",
    "Priya S","Pierre L","Rahul P","David T","Priya S"],
    "Product":["Laptop","Laptop","Laptop","Laptop","Laptop",
    "Mobile","Ordinateur Portable","Mobile","Laptop","Smartphone"],
    "Sales":[50000,55000,1200,1200,1500,None,1300,9999999,1100,35000],
    "Currency":["INR","INR","EUR","EUR","USD","INR","EUR","INR","GBP","INR"],
    "Rating":[4,5,4,4,6,3,4,5,2,0]
})

df=df.drop_duplicates()

df.SalesPerson=df.SalesPerson.replace({"Rajkumar":"Raj Kumar"})
df.Product=df.Product.replace({"Ordinateur Portable":"Laptop"})

df.loc[(df.Rating<1)|(df.Rating>5),"Rating"]=None
df.Rating=df.Rating.fillna(df.Rating.median())
df.Sales=df.Sales.fillna(df.Sales.median())

rate={"INR":0.012,"USD":1,"EUR":1.09,"GBP":1.27}
df["SalesUSD"]=df.Sales*df.Currency.map(rate)

print(df)
print("\nRegional Sales Ranking:")
print(df.groupby("Region").SalesUSD.sum().sort_values(ascending=False))
```

Output:

```
...      Region SalesPerson  Product    Sales  Currency  Rating  \
0        Asia    Raj Kumar   Laptop    50000.0    INR      4.0
1        Asia    Raj Kumar   Laptop    55000.0    INR      5.0
2        Europe   Anna M     Laptop     1200.0    EUR      4.0
4  North America   John D     Laptop     1500.0    USD      4.0
5        Asia    Priya S     Mobile    18250.0    INR      3.0
6        Europe  Pierre L     Laptop     1300.0    EUR      4.0
7        Asia    Rahul P     Mobile   9999999.0    INR      5.0
8         UK     David T     Laptop     1100.0    GBP      2.0
9        Asia    Priya S     Smartphone  35000.0    INR      4.0
```

```
      SalesUSD
0      600.000
1      660.000
2     1308.000
4     1500.000
5      219.000
6     1417.000
7  119999.988
8     1397.000
9      420.000
```

Regional Sales Ranking:

Region

Asia 121898.988

Europe 2725.000

North America 1500.000

UK 1397.000

Name: SalesUSD, dtype: float64